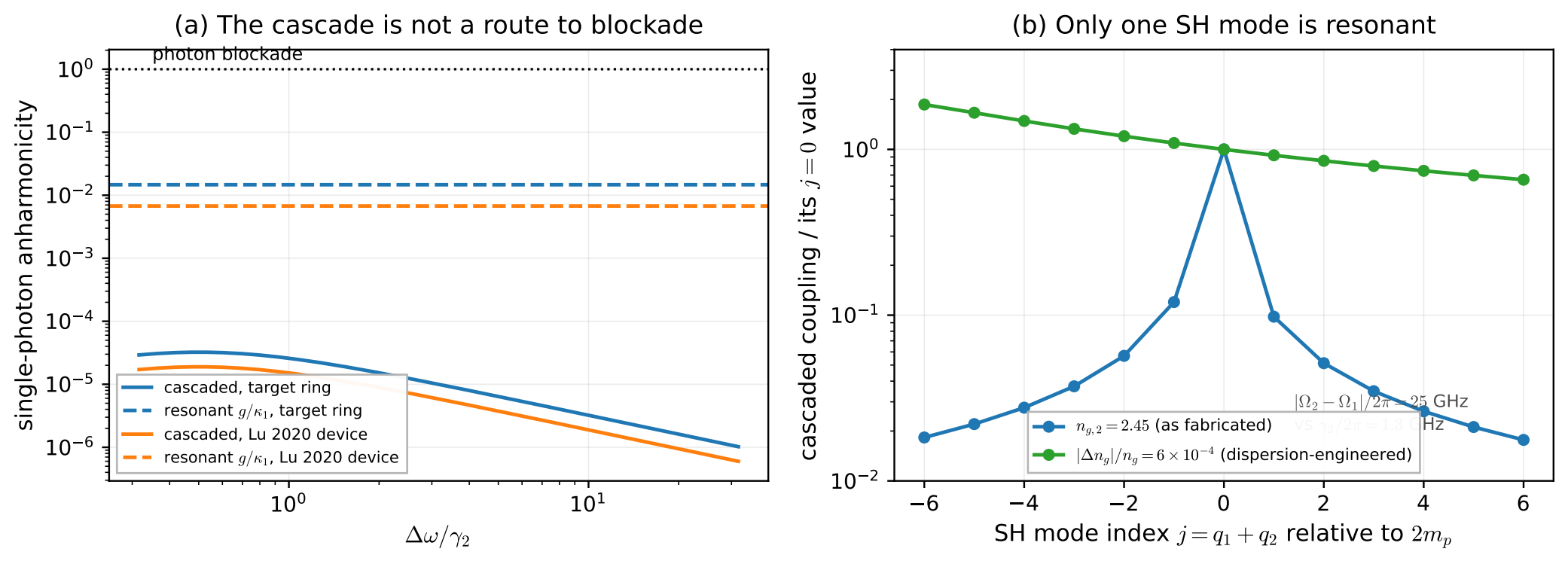
(a) The cascade is not a route to blockade
photon blockade
single-photon anharmonicity
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
10^{-6}
cascaded, target ring
resonant g/κ_1, target ring
cascaded, Lu 2020 device
resonant g/κ_1, Lu 2020 device
10^0
10^1
$\Delta\omega/\gamma_2$
(b) Only one SH mode is resonant
cascaded coupling / its $j=0$ value
10^0
10^{-1}
10^{-2}
$|\Omega_2-\Omega_1|/2\pi = 25$ GHz
vs $\gamma_2/2\pi = 1.3$ GHz
$n_{g,2}=2.45$ (as fabricated)
$|\Delta n_g|/n_g = 6\times10^{-4}$ (dispersion-engineered)
−6
−4
−2
0
2
4
6
SH mode index $j=q_1+q_2$ relative to $2m_p$